

01 / Explain the effect clearly

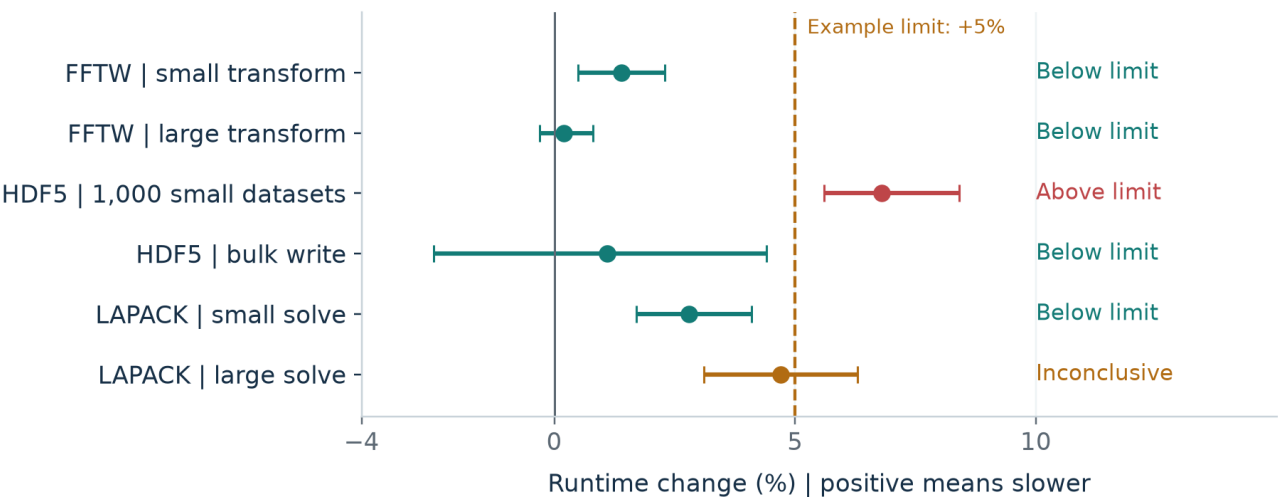
GCC 12.5.0 hardening trial | FFTW, HDF5 and LAPACK | Presentation draft

All chart values and intervals are invented examples. They are not expected overhead, measured results, or a security-compliance finding. The +5% limit is illustrative only.

The question: does full hardening change elapsed runtime enough to matter for the tested workload and scale? Start with a workload comparison, then show scale and the individual control experiments.

Do the measured workloads meet the agreed limit?

Full hardening versus optimized reference | library scope | one node | primary phases



ILLUSTRATIVE DATA | Not measured results | Example limit, not an approved policy

Read the figure: the dot is the paired runtime-change estimate; the horizontal bar is the 95% bootstrap interval. Zero means equal runtime. Positive values mean the hardened build took longer.

Interval relative to the agreed limit	Statement for the audience
Upper endpoint at or below the limit	The recorded workload meets the limit under the tested conditions.
Lower endpoint above the limit	The recorded workload exceeds the limit; repeat and investigate.
Interval crosses the limit	The result is inconclusive; improve precision or repeat the allocation.

A small detectable change may still meet the limit. An interval containing zero does not prove zero overhead. Each workload needs its own recorded conditions.

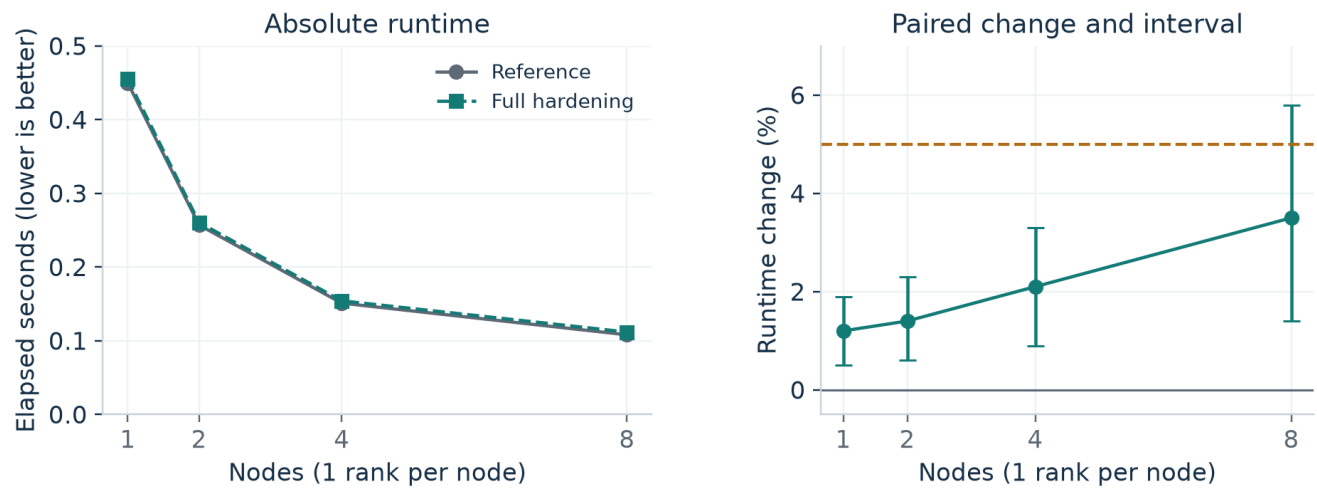
02 / Show how the effect scales

Report the absolute time and the relative cost together

One-node measurements can expose compiler overhead. Multi-node measurements test whether communication, synchronization or storage changes the observed effect. More nodes are not required for a cost to exist.

Keep scale visible when comparing performance

FFTW MPI | fixed global transform | strong scaling | maximum-rank execution time



ILLUSTRATIVE DATA | Not measured results | Example limit, not an approved policy

Reading this example: elapsed time falls as nodes increase, while the relative full/reference difference grows. At eight nodes, the interval crosses the example +5% limit. The curves illustrate a possible pattern, not a claim about FFTW.

Experiment	Keep explicit
Strong scaling	Same global FFT or HDF5 workload; show nodes, total ranks and ranks per node.
Weak scaling	HDF5 only in this campaign: same data per rank; total data grows. Use a separate plot.
Storage and threading	Separate local and shared storage; separate one-thread and four-thread FFTW cases.
Timing scope	Label planning, execution, I/O or process elapsed; never combine their percentages.

Completion comes first. Verify Slurm completion, correctness checks, actual MPI placement and the planned matrix. Missing or failed cases need a visible status; a missing result is not a 0% change.

Real figures use the campaign CSV plus job/accounting and node-inventory records. Keep repeated allocations separate; a within-allocation interval does not establish between-allocation reproducibility.

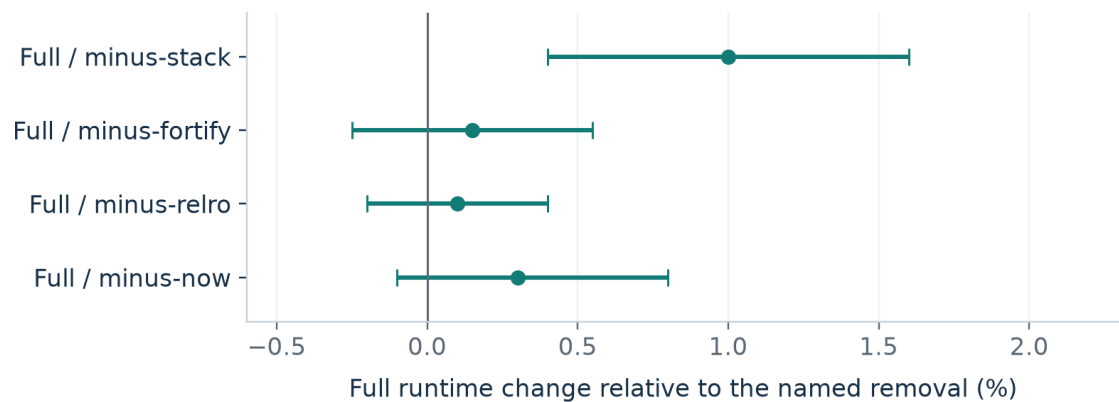
03 / Explain each control

Keep full/reference conclusions separate from control attribution

The headline compares the full library profile with the optimized reference while holding the benchmark caller fixed. The removal experiments ask what changes when one listed control is removed from that profile.

Which control merits a closer look?

FFTW small transform | same full caller | library controls removed one at a time



These contrasts are descriptive. Do not add the percentages together.

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Comparison	What it can explain
Full / reference	Combined library-profile runtime effect for a named workload.
Full / minus-stack or minus-fortify	Potential cost of those library controls in that specific build and workload.
Full / minus-relro or minus-now	Inspect short process_elapsed cases as well as warmed kernel phases.
PIE / non-PIE caller	A separate executable experiment; both arms use the full libraries.

Three boundaries: removal effects can interact, so they cannot be summed; Fortran LAPACK has no FORTIFY contrast in this matrix; PIE is not a shared-library treatment.

Performance figures answer a runtime question. Build logs and ELF/artifact checks show which protections were applied. Low measured overhead does not demonstrate that a protection is ineffective or establish STIG compliance.

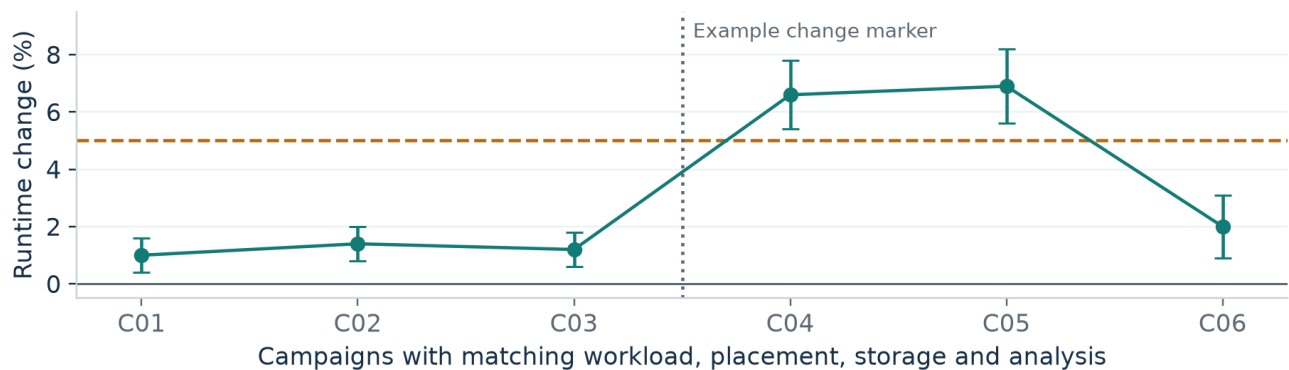
04 / History and display options

A meeting report now; a maintained dashboard when repeated monitoring is needed

ReFrame already produces JSON reports and performance logs. Its database-backed history and query/comparison commands arrived in 4.7, released in November 2024; the FOSDEM 2025 talk described this work. Our pinned 4.10.4 includes it. [1-3]

Has the same workload changed across campaigns?

Synthetic history | one point per campaign with a within-campaign interval



ILLUSTRATIVE DATA | Not measured results | Example limit, not an approved policy

Use matching conditions. A line across campaigns is meaningful only when workload, treatment, compiler/dependencies, node class, rank layout, storage and analysis match. A change marker records a candidate explanation; it does not establish causation.

Presentation need	Recommended data route
This finite trial / ISSM meeting	phases.csv + raw evidence -> static plots -> this PDF or slides.
Native ReFrame history	Enable SQLite storage before runs; use stored-session queries and comparisons.
Ongoing Grafana monitoring	Validated campaign records -> maintained PostgreSQL or Elasticsearch -> Grafana.

ReFrame documents an httpjson performance-log handler. Grafana can query Elasticsearch or PostgreSQL, but a receiver/importer, data mapping and dashboard are additional components. Preserve the paired estimate and both interval endpoints; native time aggregation must not replace that analysis. [1,4]

Future dashboard panels: matrix completion and failures; change plus uncertainty; absolute runtime; comparable campaign history; links to raw samples and build identity.

05 / Roadmap and useful additions

Separate upstream capabilities from our proposed analysis tooling

As checked on 2026-10-09, 4.10.4 is the newest stable release. The public planning signals are issues, pull requests and milestones; no separate maintained feature roadmap was found. Open milestones include 4.10.5, 4.11 and 5.0. Target dates and assigned issues express intent, not guaranteed delivery. [5,6]

Already present	Useful additions for this trial
ReFrame metric extraction, thresholds, JSON, performance logs, SQLite history and queries	An immutable, versioned export joining paired results to build/source identities.
Paired harness, raw samples, bootstrap intervals, correctness and placement checks	A final expected-matrix completeness check with explicit failed/missing/not-applicable states.
Campaign labels, node inventory and per-phase CSV	Comparability guards, independent allocation replication, and a one-command report snapshot.

Possible generic upstream proposals: richer typed provenance in exports; uncertainty-aware comparison hooks; baseline comparability checks; portable plot/report templates. Start with a concrete reproducer and a maintainer discussion. These are our suggestions, not announced ReFrame commitments.

A five-minute meeting sequence

1. State the question and the exact flag/build scope. **2.** Confirm correctness and coverage. **3.** Show change and intervals against the agreed limit. **4.** Show scale and the startup/control contrasts. **5.** Close with bounded conclusions and unresolved cases.

Example conclusion once real data exists: "For [workload, scale and storage], full hardening changed runtime by [estimate]% with a 95% interval of [low, high]%. This [meets / exceeds / does not resolve] the predeclared [limit]% limit."

Primary sources and the full implementation plan

- [1] ReFrame reporting, HTTP logs and history
 - [2] ReFrame storage, queries and session metadata
 - [3] FOSDEM 2025 performance analytics talk
 - [4] Grafana Elasticsearch / PostgreSQL data sources
 - [5] ReFrame v4.10.4 release
 - [6] ReFrame milestones and tracked work
- Trial results presentation plan and reusable figure files